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Corruption In Health Care Systems: Trends In Informal Payments Across Twenty-Eight EU Countries, 2013–19

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ABSTRACT Corruption is a major challenge in health care systems across the European Union (EU), where it manifests most visibly as informal payments from patients to providers. A higher prevalence of informal payments has been associated with lower public health care expenditure. EU member states have experienced significant changes in public health care expenditure throughout the 2000s. Given the lack of research on the topic, we explored trends in informal payments using representative data from twenty-eight EU member states during the period 2013–19 and in relation to changes in public health care expenditure. Overall, we found that informal payments increased in 2019 compared with 2013, whereas the perception of corruption decreased. Although higher public health care expenditure was associated with less corruption, we found a smaller effect size between informal payments and this expenditure throughout the study period. Our results suggest that informal payments may be driven by other factors, although the directionality of this relationship requires further investigation. Moreover, additional public health care investments may be insufficient to confront corruption unless coupled with measures to limit wasteful spending and increase transparency. Policy makers should understand that factors external to health systems, including media coverage and cultural and political factors, should be explored to explain country-level differences in corruption.

Health care systems are increasingly confronted with challenges as they try to meet population health needs, slowing progress toward the achievement of universal health coverage. Inadequate funding and the waste of resources often experienced by health systems can be further exacerbated by corruption. Defined as "the abuse of entrusted power for private gain,"[1] corruption is a complex phenomenon that is tightly linked to social, political, and economic factors.[2] Uncertainty, asymmetry of information, and large numbers of actors make health systems vulnerable to corruption, which may manifest in various forms including informal payments, absenteeism, reimbursement fraud, misuse of dual practice [health professionals working simultaneously in the public and private sectors], abuse of high-level positions, and embezzlement of drugs or medical devices.[3,4] Corruption limits access to health care services, increases health inequalities, and worsens health outcomes.[4–7] At the global level, United Nations Sustainable Development Goal 16.5 calls for reducing all forms of bribery and corruption by 2030.[8,9]

It was estimated in 2011 that about 56 billion euros are lost annually in the European Union (EU) because of corruption in health care, although the impact of this corruption on individ-

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uals and society as a whole may largely exceed the monetary value of its consequences.[10–12] In Europe, higher levels of health care corruption have been associated with an increasing risk of the population developing chronic diseases[13] and antibiotic resistance,[14,15] lower trust in public health care facilities,[16] and lower perception of quality of health care services,[17] as well as inefficient allocation of health budgets and increased health inequalities.[10]

Two studies conducted as part of an anticorruption strategy set up by the European Commission in 2011 found that corruption was most prevalent in Southern and Eastern Europe, least common in Northern Europe, and less widespread in Western Europe.[10,18] Informal payments—defined as a transfer of money from patients to providers that was either offered by patients or demanded by physicians in exchange for preferential treatment—were the most visible manifestation of corruption, particularly in Eastern European countries.[18] Both studies found that systemic issues in health systems, especially low health care funding, were the main drivers of informal payments.[10,18]

Several studies investigated the burden of informal payments in health service sdelivery, both at a single-country level[19–22] and a cross-national level.[23–28] The majority of studies focus on Eastern Europe, where informal payments are widespread because of low health care funds, poor governance, barriers to accessing health care services, and high levels of out-of-pocket payment.[10,29] The lack of consistency in methodology can generate contrasting results, making it difficult to generalize the magnitude, costs, and consequences of corruption across countries and over time.[9,23–27,29,30] A 2018 systematic review of thirty-eight studies on informal payments from eighteen countries found that the prevalence of informal payments ranged between 2 percent and 80 percent across studies, which differed extensively in methodology.[29] In a study comparing data from six Eastern European countries, the prevalence of informal payments varied significantly across countries, being higher when the recall period was extended from twelve months to infinity.[24] A recent study analyzing trends in informal payments in EU health care systems during 2013–19 found that although there was no change at the European level, the prevalence of informal payments increased in the majority of countries but decreased in Romania, Lithuania, and Sweden throughout the study period.[31]

Cross-European studies have found a strong negative relationship between levels of health care expenditure and informal payments in EU health care systems.[32,33] The highest levels of informal payments were observed in countries with the lowest public health care expenditure, including the majority of Eastern European countries and some Southern European countries. Conversely, informal payments were least common in countries with the highest levels of public health care expenditure, including Denmark and the United Kingdom (in studies conducted while the UK was still a member of the EU).[32,33] Health budget allocation varies significantly across EU member states, depending on both fiscal capacity and political priority.[34] The growth in public health care expenditure observed between 2000 and 2007 was reversed after the 2008 financial crisis.[34] The adoption of austerity measures to contain public spending in the years after the crisis, which mainly consisted of cuts to health budgets and coverage restrictions, resulted in a significant decline in public health care expenditure and in the growth of out-of-pocket payment across several EU member states, decreasing financial protection and increasing unmet need for health care services.[35] The influence of variations in public health care expenditure on corruption levels across EU health care systems has not been investigated.

The practice of informal payments is a threat to health system performance and the health of the population. A previous study analyzed trends in informal payments across EU member states over time,[31] but to our knowledge, no previous study has examined whether changes in the prevalence of this phenomenon are associated with variations in public health care expenditure. Using data from Special Eurobarometer surveys collected between 2013 and 2019, we investigated trends in informal payments in public health care facilities and in the perception of corruption in health care across twenty-eight EU member states over time and in relation to changes in public health care expenditure.

Study Data And Methods

STUDY POPULATION Data for our analysis were obtained from the three latest Eurobarometer surveys on corruption: Special Eurobarometer waves 79.1 (February–March 2013), 88.2 (October 2017), and 92.4 (December 2019). The survey questions and methodology were consistent across the three waves.[36–38] Respondents were people ages fifteen and older residing in a country that was a member of the EU at the time of the survey ($N = 27{,}786$ in 2013, 28,080 in 2017, and 27,498 in 2019 in twenty-eight EU member states). Samples were independent across waves. Data were collected through face-to-face interviews that were conducted in the homes of respondents and in the local language. Euro-

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barometer surveys employ multistage random probability sampling.[39] Primary sampling units were randomly selected from the administrative regions of each country and were proportional to the population size and density of that country. Within each primary sampling unit, a cluster of households was obtained by following a standard random route procedure. In each household, the household member whose birthday was closest to the date of the interview was chosen. Although response rates were not published, poststratification and population size weights were provided in the official Eurobarometer survey data sets to ensure that samples were nationally representative in terms of sex, age, and area of residence. Because Eurobarometer surveys are publicly available and data are completely anonymized, no ethical approval was required.

The overall increase in informal payments observed in our study may be partly explained by the widespread adoption of austerity measures.

MEASURES

▶ **CORRUPTION IN HEALTH CARE:** We measured corruption in health care both as the use of informal payments to health care practitioners in public health care facilities and as perceived levels of corruption in health care among respondents. We conceptualized both outcomes as binary variables for the regression models to follow.

Respondents were asked whether they had attended public health care facilities in the twelve months before survey interview, such as a public health care practitioner or a public health care clinic. Those who answered "yes" were asked the following question: "Apart from official fees, did you have to give an extra payment or a valuable gift to a nurse or a doctor, or make a donation to the hospital?" Response options were "yes," "no," and "don't know." To estimate the prevalence of informal payments in the overall study population, we considered respondents who answered "yes."

To assess the perception of corruption in health care, all respondents were asked the following question: "In [our country], do you think that the giving and taking of bribes and the abuse of power for personal gain are widespread among any of the following?" The interviewer then listed several authorities and institutions, including the health care system. For each item, response options were "yes," "no," and "don't know." To assess the perception of corruption in health care in the overall study population, we considered respondents who answered "yes" in relation to the health care system.

▶ **SOCIODEMOGRAPHIC DATA:** At the individual level, we included the following categorical covariates based on self-reported data from respondents: age (15–24, 25–39, 40–54, or 55 and older), gender (male or female), marital status (married, remarried, or single with a partner; unmarried; or divorced, separated, or widowed), age on completion of full-time education (no full-time education; age 15 or younger, ages 16–19, or age 20 or older; or still studying), area of residence (rural area or village, small or middle-size town, or large town), and difficulty paying bills in the twelve months before the interview (never or almost never, most of the time, or from time to time).

▶ **COUNTRY-LEVEL DATA:** At the country level, we obtained data about public health care expenditure as a percentage of gross domestic product (GDP) comprising spending by government and compulsory contributory health care financing schemes from Eurostat[40] for twenty-eight EU member states for each survey year. Because 2019 data for public health care expenditure were not available for several EU member states, we used 2018 data for all countries. Data for public health care expenditure for Malta and Slovenia in 2013 were missing, so we used data from 2014.

To control for differences in wealth between countries, we obtained data about GDP per capita in purchasing power standard from Eurostat for twenty-eight EU member states for each survey year, expressed as a continuous variable with 1,000 purchasing power standard increments. The purchasing power standard is an artificial currency unit used by Eurostat for cross-national comparisons that accounts for differences in price levels between countries.[41]

STATISTICAL ANALYSIS Results from descriptive analyses are presented as weighted percentages with 95% confidence intervals, whereas results from regression analysis are presented as adjusted prevalence ratios with 95% confidence intervals.

Mixed-effects two-level Poisson regression models with random intercepts were fit in the

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Our findings point toward the need to couple public health care expenditure investments with measures to strengthen health systems.

study sample with country set as random intercept. A total of 80,082 respondents across three survey waves nested within twenty-eight countries were included in regression analysis; those with missing data were excluded. We built two models for each binary outcome variable: informal payments and the perception of corruption in health care. We used Akaike and Bayesian information criteria as measures to evaluate fitness during model building. Model 1 assessed the association between time (independent variable) and each of the outcomes, adjusted for age, gender, marital status, age on education completion, area of residence, difficulty paying bills, and year of survey (categorical variable). Model 2 assessed the relationship between public health care expenditure (independent variable) and each of the outcomes, adjusted for the same individual-level variables and GDP per capita. We included an interaction term between time and public health care expenditure in model 2 to explore whether the relationship between public health care expenditure and informal payments, or perception of corruption in health care, varied over time. To interpret the interaction term, we plotted differences in predicted probabilities for informal payments or perception of corruption between survey years over the range of values of public health care expenditure observed across EU member states.

We built a single-level Poisson regression model with the same covariates as model 1 for each country to explore the association between informal payments, or perception of corruption in health care, and time in individual EU member states. Analyses were performed with Stata/IC, version 15.1.

LIMITATIONS The consistency in methodology of Eurobarometer surveys allowed us to systematically compare trends in informal payments across EU health care systems over time in nationally representative samples. The analyses relied on self-reported data; hence, it is plausible that the prevalence of informal payments was under- or overestimated.[30] Both measures of informal payments and perception of corruption were included in this study to account for the discrepancy that is usually reported between them.[11] Any inconsistencies in results were likely to be equal across countries and over time. Given the cross-sectional nature of this study, it was not possible to establish causality in the relationships between informal payments, perception of corruption in health care, and public health care expenditure. Unlike previous studies, the definition of informal payments in Eurobarometer surveys did not distinguish between under-the-table payments and gifts, which are a common action of gratitude in some countries and might not be considered a corrupt practice.[42] The twelve-month recall period may have increased risk for bias, given that Eurobarometer surveys do not distinguish between the use of informal payments for inpatient and outpatient services. However, evidence shows that informal payments are more prevalent in inpatient services, for which a twelve-month recall period is commonly used.[29]

Study Results

A total of 83,364 citizens from twenty-eight EU member states were surveyed across the three Eurobarometer waves (online appendix exhibit A1).[43] The regression analysis was performed with 80,082 respondents.

In the EU, the prevalence of informal payments from patients to health services providers or health care institutions was 3.6 percent in 2013, 3.2 percent in 2017, and 3.8 percent in 2019, although figures varied considerably at the country level (exhibit 1, appendix exhibit A2),[43] in 2013 ranging from 0.2 percent in Finland to 15.8 percent in Lithuania, and in 2019 ranging from 0.0 percent in Sweden to 11.8 percent in Austria. The divergence between Northern and Eastern European countries remained consistent throughout the study period, although Romania and Lithuania had lower prevalence of informal payments in 2019 than in 2013. Informal payments were considerably more common in Western European countries in 2019 than in 2013.

In the EU, the proportion of people who perceived that corruption in the health care system as widespread was 33.6 percent in 2013, 31.0 percent in 2017, and 26.5 percent in 2019, with some differences at the country level (exhibit 2,

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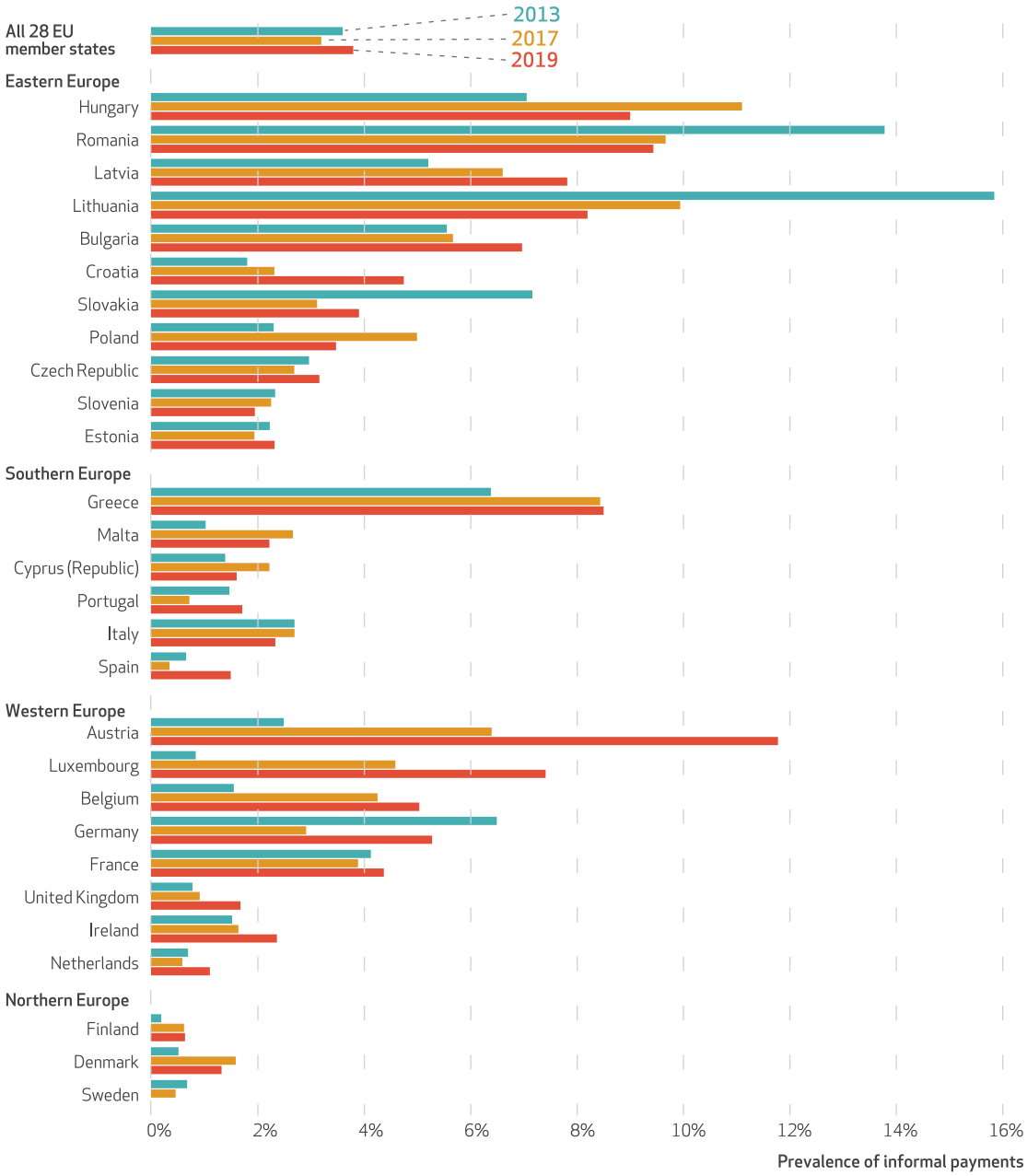
EXHIBIT 1

Trends in the prevalence of informal payments in public health care facilities, using representative data from 28 European Union (EU) member states, by European subregion, selected years 2013–19



SOURCE Authors' analysis of data from Eurobarometer surveys, 2013, 2017, and 2019.

appendix exhibit A3).[43] The biggest difference was observed between Eastern and Northern European countries throughout the study period, although the majority of Eastern European countries showed lower levels of perception of corruption in 2019 than in 2013. Sweden was the only Northern European country to observe higher perceived levels of corruption in 2019 than in 2013. Greece was the EU member state with the highest perception of corruption throughout the study period (81.3 percent in 2013, 81.0 percent in 2017, and 80.6 percent in 2019), whereas Finland recorded the lowest prevalence throughout the same period (3.9 percent in 2013, 5.0 percent in 2017, and 3.8 percent in 2019).

After sociodemographic factors were adjusted for, the prevalence of informal payments in EU

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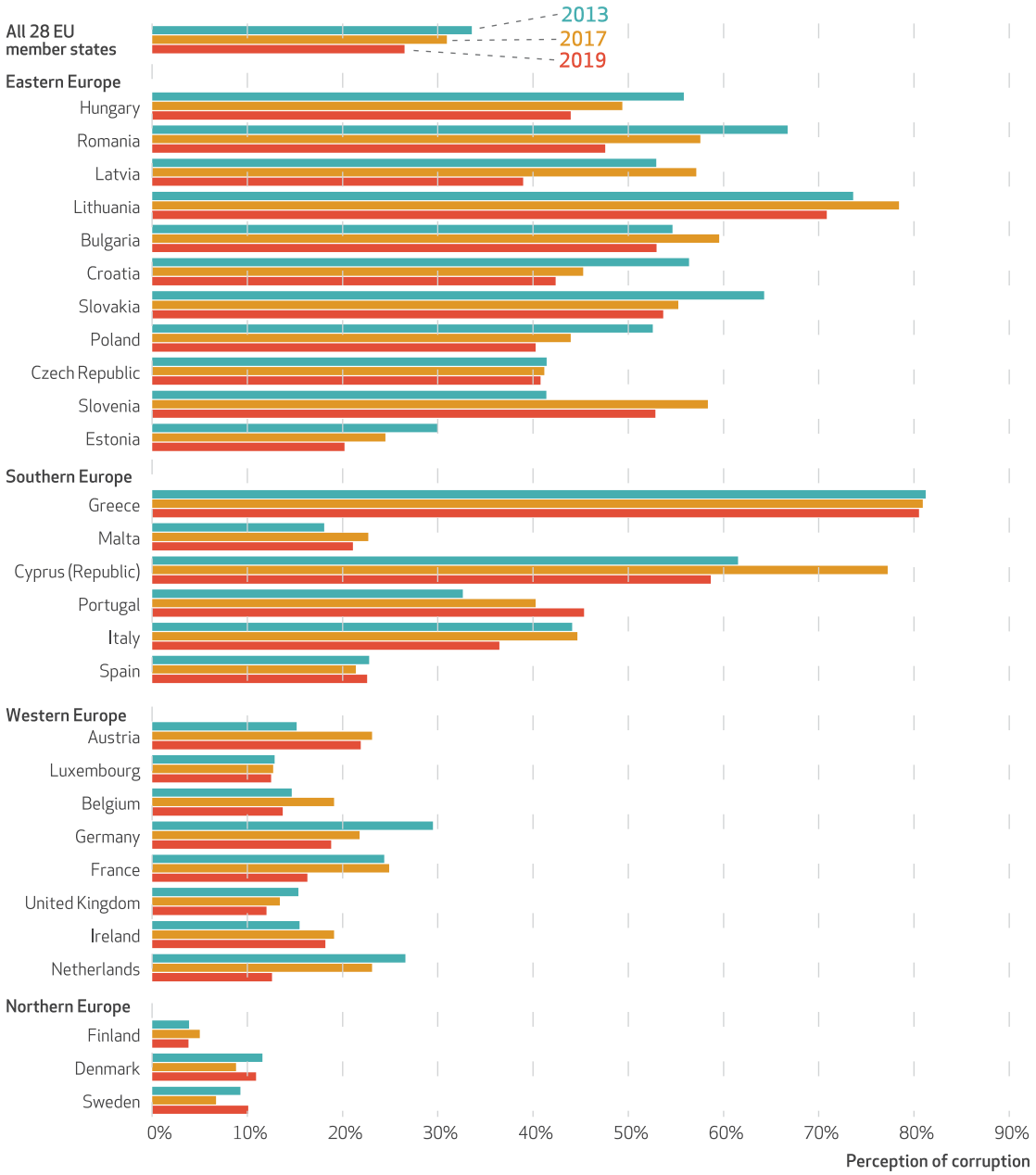
EXHIBIT 2

Trends in the proportion of respondents who perceived widespread corruption in their country's health care system, using representative data from 28 European Union (EU) member states, by European subregion, selected years 2013–19



SOURCE Authors' analysis of data from Eurobarometer surveys, 2013, 2017, and 2019.

health care systems was 14 percent higher in 2019 than in 2013 (aPR: 1.14; 95% CI: 1.05, 1.25) (exhibit 3, model 1). After the same individual-level variables were adjusted for, prevalence increased statistically significantly in seven EU member states (Austria, Belgium, Bulgaria, Croatia, Denmark, Latvia, and Luxembourg) and decreased in four (Germany, Lithuania, Romania, and Slovakia) (exhibit 4, appendix exhibit A4).[43]

Results from model 2 (exhibit 3) show that in 2013 the prevalence of informal payments was 25 percent lower for every additional percentage point of GDP invested in public health care expenditure (aPR: 0.75; 95% CI: 0.66, 0.86) after adjustment for individual covariates and GDP per capita. The association between public health care expenditure and informal payments differed over time (aPR: 1.07; 95% CI: 1.02, 1.12 for the interaction term), with countries spend-

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EXHIBIT 3

Associations between time and expenditure variables with informal payments and perception of corruption in health care across 28 European Union member states, selected years 2013–19

	Adjusted prevalence ratio			
	Informal payments		Perception of corruption	
Variables	Model 1[a]	Model 2[b]	Model 1[a]	Model 2[b]
Time				
2013	1.00	1.00	1.00	1.00
2017	1.01	1.04	1.00	1.09
2019	1.14**	0.79	0.89**	1.04
Public health care expenditure (% of GDP)	—[c]	0.75**	—[c]	0.84**
GDP per capita in PPS (1,000 increments)	—[c]	1.00	—[c]	0.99
Interaction between time and public health care expenditure				
2013	—[c]	1.00	—[c]	1.00
2017	—[c]	0.99	—[c]	0.98
2019	—[c]	1.07**	—[c]	0.97**

SOURCE Authors' analysis of data from Eurobarometer surveys, 2013, 2017, and 2019. **NOTES** The table shows the results of Poisson multilevel regression models ($N = 80{,}082$). GDP is gross domestic product. PPS is purchasing power standard. [a]Model 1 adjusted for time, age, gender, marital status, age on education completion, area of residence, and difficulty paying bills. [b]Model 2 adjusted for public health care expenditure and GDP per capita in addition to the variables in model 1. [c]The corresponding variable was not included in the model. $^{**}p < 0.05$

ing 5.5 percent or more of GDP on public health care expenditure reporting higher prevalence of informal payments in 2019 than in 2013 (appendix exhibit A5).[43] The relationship between informal payments and GDP per capita was not statistically significant (aPR,: 1.00; 95% CI: 0.98, 1.02).

Although the perception of corruption in the EU in 2017 did not differ significantly from that in 2013 (aPR: 1.00; 95% CI: 0.97, 1.03), it decreased by 11 percent in 2019 compared with 2013 (aPR: 0.89; 95% CI: 0.87, 0.92), after adjustment for sociodemographic factors (exhibit 3, model 1). There was wide variation in perception of corruption at the country level between 2013 and 2019, ranging from a 47 percent decline in the Netherlands (aPR: 0.53; 95% CI: 0.43, 0.66) to a 32 percent increase (aPR: 1.32) in both Ireland (95% CI: 1.05, 1.65) and Austria (95% CI: 1.06, 1.64) (appendix exhibits A4 and A6).[43]

Results from model 2 (exhibit 3) show that in 2013 every additional percentage point of GDP invested in public health care expenditure was associated with a 16 percent decrease in the perception of corruption (aPR: 0.84; 95% CI: 0.79, 0.88) after adjustment for individual covariates and GDP per capita. The association between public health care expenditure and the perception of corruption in health care was different across times (interaction term: aPR: 0.97; 95% CI: 0.96, 0.99). Specifically, the perception of corruption was significantly lower in 2019 than in 2013 for public health care expenditure at or above 3.8 percent of GDP (appendix exhibit A7).[43] The relationship between the perception of corruption in health care and GDP per capita was not statistically significant (aPR: 0.99; 95% CI: 0.98, 1.00).

Discussion

Our study found that the prevalence of informal payments in EU health care systems increased between 2013 and 2019, whereas the perception of corruption in health care decreased, with substantial variation at the country level. Higher levels of public health care expenditure were associated with lower prevalence of informal payments and perception of corruption, although this relationship varied during the study period. In 2019 countries with higher public health care expenditure observed higher prevalence of informal payments but lower perception of corruption compared with 2013.

Our finding that informal payments were more common in EU member states that invested lower shares of GDP in public health care expenditure is consistent with the findings of previous cross-sectional studies showing that informal payments were significantly more common in European countries with lower public health care expenditure.[32,33] Although there is insufficient evidence to explain the directionality of this relationship, higher levels of public health care expenditure may contribute to cov-

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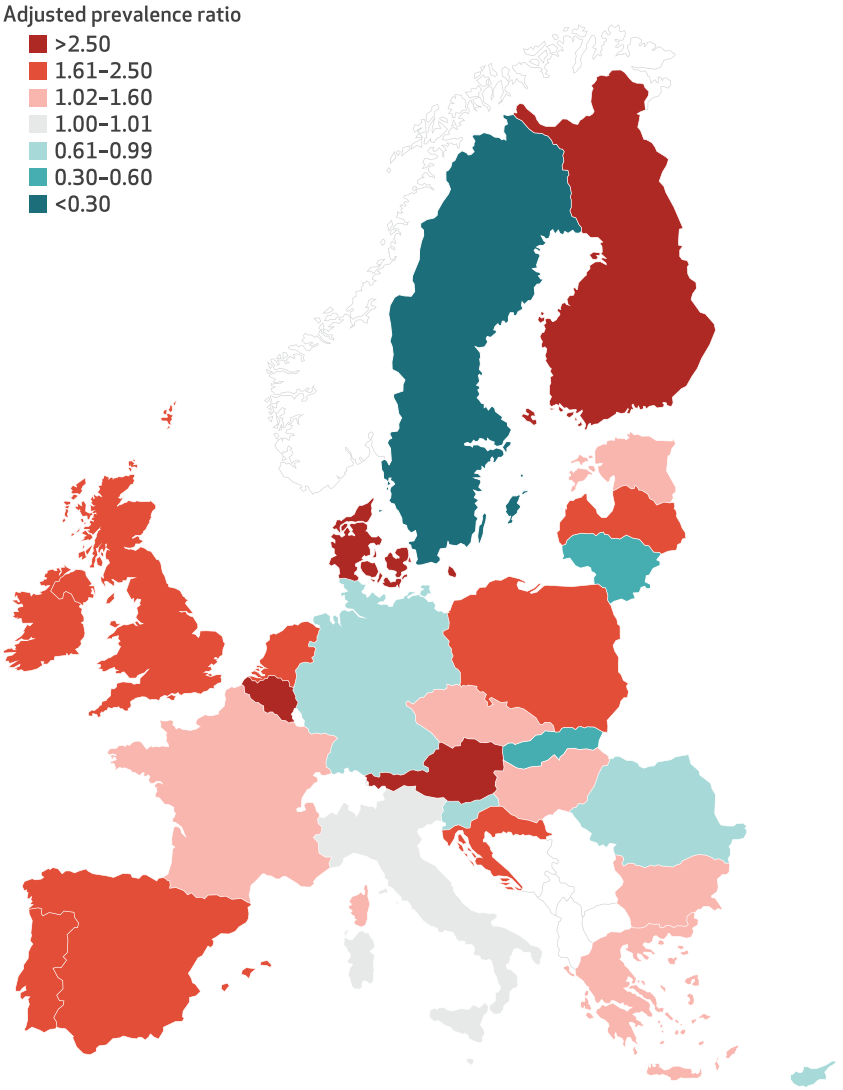
EXHIBIT 4

Changes in informal payments across 28 European Union member states from 2013 to 2019



SOURCE Authors' analysis of data from Eurobarometer surveys, 2013, 2017, and 2019. **NOTES** $N = 80{,}082$. Changes are expressed as adjusted prevalence ratio for 2019 (reference = 2013). Shades of red represent increases in prevalence of informal payments throughout the study period, whereas shades of blue represent decreases. Light gray represents negligible or no change in the prevalence of informal payments. Countries that are white are not in the European Union and were not studied. The data include the United Kingdom before its departure from the EU became official in 2020.

ering costs and increasing the availability of resources, especially in underfunded health care systems characterized by poor infrastructure, a limited range of services, and low salaries for health care staff, thereby reducing the incentives for providers to ask for bribes and the need for the public to pay informally.[33]

The inverse relationship between public health care expenditure and informal payments became less strong during the study period in countries with higher levels of public health care expenditure, potentially suggesting that additional public health care investments may have been insufficient to further reduce the prevalence of informal payments. To the best of our knowledge, no previous study has examined trends in informal payments over time at different levels of public health care expenditure. Health systems across the EU have adopted measures to tackle inefficiencies after increasing fiscal pressures resulting from the 2008 financial crisis, the demands of aging populations, and new investments in medical technologies.[44–46] It can be argued that the austerity measures adopted by EU member states after the financial crisis, including cuts to health budgets or restric-

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tions on health care coverage, may have encouraged the use of informal payments, consequently reducing access, public satisfaction, and transparency within health systems and increasing the risk for catastrophic health care expenditure or delayed treatment seeking for patients who could not afford to pay informally or for private health care.[35,44,47,48] We hypothesize that the overall increase in informal payments observed in our study may be partly explained by the widespread adoption of austerity measures. Germany was an exception, in that it opted for financial stimulus after the crisis and abolished cost sharing for outpatient visits in 2012 and saw a decrease in the prevalence of informal payments between 2013 and 2019, according to our study.[35,45] However, this hypothesis requires further investigation.

Although our study observed an increase in the prevalence of informal payments in EU health care systems between 2013 and 2019, the perception of corruption in health care decreased during this period. Periods of economic recession have been associated with higher perceived corruption.[49] Thus, this apparent decrease may represent a regression to the mean, given that 2013 figures could have been inflated by the 2008 financial crisis. Public perception of corruption may also be heavily influenced by the media, as greater coverage of corruption episodes has been associated with an increase in the perception of corruption, although this may vary depending on media content.[18] In general, the media prioritizes scandals, which attract the attention of the public and might be related to parts of the health care system that patients have not come into contact with, whereas smaller-scale episodes of corruption such as informal payments often remain underreported.[11] As a consequence, the public may conceptualize corruption in a way that does not necessarily capture the frequency of informal payments,[50] which may often be seen as a means to express gratitude, not a corrupt practice.[42]

Our study showed that higher levels of public health care expenditure were associated with lower perception of corruption in health care and that this relationship was stronger in 2019 than in 2013. These findings can be related to studies that found higher satisfaction with and better perception of safety in health care systems in European countries with higher levels of health care expenditure. Higher investments in health care may result in greater availability of resources, higher accessibility and transparency, shorter waiting times, and lower costs for patients, and consequently better public perception of health care systems, which is also influenced by perceived levels of corruption.[51,52]

The practice of informal payments poses significant challenges for the achievement of universal health coverage in the EU.

Policy Implications

Our findings can inform further research on informal payments that should consider using different recall periods, distinguishing between types of services for which patients pay informally, or assessing the use of other methods to access health care services such as personal connections. Future research should also investigate drivers of informal payments at the health systems level, in relation to their structure, financing, and provider payment schemes, to increase understanding of the directionality of the relationship between informal payments and public health care expenditure. This should be accompanied by research toward the development of standardized measures of corruption to improve the quality of evidence.

Our findings point toward the need to couple public health care expenditure investments across EU member states with measures to strengthen health systems, such as monitoring the use of resources to prevent inefficient spending of health care funds and increasing transparency. Further, health care corruption is a multifaceted phenomenon that has often been linked to factors external to health care systems, including media coverage as well as cultural and political factors; this suggests that research on this topic should also consider the wider context to develop effective anticorruption strategies and ultimately improve the health of the population.

Conclusion

The practice of informal payments poses significant challenges for the achievement of universal health coverage in the EU. This was the first study to examine trends in informal payments in health care systems between 2013 and 2019, using representative data from twenty-eight EU member states, in relation to changes in public

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health care expenditure. Overall, results showed that the prevalence of informal payments in EU health care systems increased, whereas the perception of corruption in health care decreased, with significant differences at the country level. The prevalence of informal payments decreased in Eastern European countries throughout the study period, whereas it increased in several Western European countries. These contextual differences may be more meaningful than overall trends when discussing policy reform. Higher public health care expenditure was associated with lower prevalence of informal payments and perception of corruption in health care, although we found a smaller effect size between informal payments and public health care expenditure during the study period. Further research is required to investigate the drivers of informal payments at the health systems level, as well as the directionality in the relationship between informal payments and public health care expenditure, given the cross-sectional nature of this study. ■

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